Kamali S

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.65
SHRI GANGA HIGHER SECONDARY SCHOOL	Class XII (STATE BOARD)	2020	93.60%
SHRI GANGA MATRIC HIGH SCHOOL	Class X (STATE BOARD)	2018	95.60%

SKILLS

Programming & Databases: Python, SQL, DBMS

Machine Learning & AI: Scikit-learn, CNN, GenAI, RAG, LangChain, Hugging Face

Data Analysis & Visualization: Pandas, NumPy, Power BI, Matplotlib, Excel

GIS & Remote Sensing: ArcGIS, QGIS, Google Earth Engine, ERDAS Imagine, SNAP, Digital Image Processing, Remote Sensing

Tools & Platforms: Git, VS Code, JIRA, Jenkins, Scrum

RELEVANT COURSES

Machine Learning & Data Analysis: Machine Learning for Process Systems Engineering, Optimization Techniques-I, Climate Data Analysis and Methods, Data Modeling in Atmospheric Sciences.

Climate Science & Analytics: Remote Sensing, GIS and Climate Change, Climate Modeling, Introduction to Climate Science and Climate Monitoring, Climate Change Mitigation and Adaptation, Climate Governance.

WORK EXPERIENCE

Full House

Onboarding Specialist (Product Analyst)

- Documented **end-to-end product functionality** across the platform's web and mobile applications, mapping each feature's workflow and backend behavior while validating performance through functional testing.
- Onboarded and registered customers and business owners onto the platform by identifying prospects, **explaining product capabilities**, and guiding them through end-to-end registration.

Hexagon Capability Center India Pvt. Ltd , Hyderabad

QA Intern

- Performed manual and ad-hoc testing on the **ERDAS IMAGINE** software to identify functional defects before release, by executing test cases across core imaging operations and workflows.
- Automated repetitive test cases to reduce manual testing effort and improve coverage, using **C#** and **WinAppDriver** to script actions like image opening and processing operations.
- Managed test cycles in a **Scrum**-based Agile environment to ensure timely delivery and traceability, using **JIRA** for tracking, **Jenkins** for builds, and virtual machines for cross-environment execution.

SAKURA Geoinformation Software Research Pvt. Ltd , Chennai

GIS Intern

- Geotagged railway assets for accurate spatial tracking by generating latitude-longitude datasets from field photographs and mapping them using Google Earth Pro and **ArcGIS** software.

- Digitized terrain features (polygons, lines, points) from ortho imagery and toposheets to build land models for Rockfield Estates, creating shapefiles and TIN/DSM surfaces.
- Mapped Adyar's urban growth from 2001 to 2019 using **Google Earth Pro** historical imagery to study urbanization trends, exporting results as ArcGIS point features.

RESEARCH

Improving S2S Precipitation Forecast Accuracy Using Deep Learning

M.Tech Thesis | Working under the guidance of Dr. Maheswaran R | IIT Hyderabad

- Collected and organised 20 years (2005-2024) of ECMWF S2S precipitation reforecasts (lead 0-46 days) over South Asia, aligned by target date and lead time. Currently running EDA of forecast vs. observed rainfall.
- Reviewed 8+ papers on ML/DL post-processing of S2S forecasts (U-Net, Attention U-Net, RF, SVM, ViT-LSTM) to identify gaps in long-lead skill, predictor selection and teleconnection use.
- Developing a U-Net model with a custom extreme-rainfall loss, atmospheric predictors and teleconnection indices (ENSO, MJO, NAO). It will be benchmarked against raw ECMWF and Quantile Mapping.

PROJECTS

Air Quality Analysis of Indian Cities

Python, SQL (SQLite), Pandas, Matplotlib

- Built an **end-to-end pipeline** on **29.5K CPCB daily records (26 cities)** to compare pollution across cities and seasons. Rule-based validation flagged an unreliable series (Ahmedabad: AQI up to 2049 vs 500 max) that had falsely ranked it the "most polluted" city.
- Isolated the 2020 COVID-lockdown effect from seasonality using **difference-in-differences** (2020 vs 2019), finding a **median 24% PM2.5 drop** across 19 cities. The naive comparison overstated Delhi's drop (-50% vs -39%), and I cross-validated results in SQL and pandas.
- Quantified seasonal and Diwali pollution patterns with SQL queries. Winter AQI was ~**2 x monsoon** in all 23 cities, and AQI was higher during Diwali in 40 of 58 city-year comparisons (median +8%).

Optimization of Rice Yield and Pesticide Usage

ANN + Multi-Objective Genetic Algorithms (NSGA-II/III)

- Engineered an ANN surrogate model (TensorFlow/Keras, 2 hidden layers) to predict rice pesticide usage and production across **1,197 records**, achieving **R2 = 0.95 (train), 0.95 (validation), 0.91 (test)** to replace costly field trials with fast simulations.
- Optimized pesticide vs. production **trade-offs** using **NSGA-II/NSGA-III** (pymoo), evolving **1**00 candidates over 100 generations to generate a Pareto-optimal solution set.
- Applied **knee-point detection & data visualization** (NumPy, Matplotlib) on the Pareto front to auto-select the optimal trade-off, delivering actionable region- and season-specific input recommendations.

Banking FAQ Chatbot Using RAG

GenAI | RAG + FAISS Vector Search

- Built a RAG chatbot with **LangChain** and **Streamlit** that answers banking queries on loans, cards, KYC and online banking, and shows the source category with each answer.
- Embedded FAQs with Hugging Face Sentence-Transformers, indexed them in **FAISS**, and generated answers with **Llama 3.1(Groq)** at low temperature, so replies stay grounded in retrieved context.
- Added a similarity-threshold guardrail that returns a fallback refusal for out-of-scope queries, **preventing hallucinated answers** in a finance setting.

POSITIONS OF RESPONSIBILITY

Placement Coordinator - Department of Climate Change, IIT Hyderabad

- Working across **Core Climate (GSS) and AI/ML Cohort '26**; pitch and onboard new and existing companies to grow the department's recruiting network.
- Identify and contact key POCs at target companies to build and sustain recruitment pipelines across both domains.

Social Media & Web Coordinator - Department of Climate Change, IIT Hyderabad

- Manage the department's LinkedIn page and website, posting updates and content to drive outreach and visibility for the Department of Climate Change.

Teaching Assistant , IIT Hyderabad

- CC5380: Paleoclimatology: A Peek into Ancient Climate Change

HOBBIES & INTERESTS

Dance, Badminton, Carom, Art